Sliding Window - List/Array Mastery Sheet

Python | Prerequisite: Hashing + Two Pointers | Basic to Advanced

1. Best Material (sirf ye kaafi hai)

• **Aditya Verma - Sliding Window playlist** (YouTube, Hindi): fixed + variable window ka concept clear hota hai.
• **NeetCode - Sliding Window section** (neetcode.io roadmap): LeetCode ke important questions, short explanations.
• **Striver A2Z / SDE sheet**: Two Pointers + Sliding Window section (practice ke liye).
• **Hello Interview - Sliding Window guide**: fixed-length aur variable-length ke templates.
• Rule: video ek baar dekho, phir seedha questions. Video me hi atke mat raho.

2. Trigger Words - Question me ye dikhe to Sliding Window socho

Words in question	Socho	Type
subarray of size K / every window of K / K consecutive	Window size fix, add 1 remove 1	Fixed
longest / max length subarray with at most K (zeros, distinct, flips, sum <= K)	Expand right, invalid hote hi left shrink	Variable - Longest
smallest / minimum length subarray with sum >= target	Valid hote hi shrink karke min nikalo	Variable - Shortest
number of subarrays with ... less than / at most K	Har right par count += r-l+1	Counting
exactly K (distinct / odd numbers / sum)	atMost(K) - atMost(K-1)	Counting - Exactly
at least K (count)	Valid milte hi count += n - r	Counting - At least
max / min of EVERY window	Monotonic Deque	Fixed + Deque
pick K cards / elements from both ends	Bahar ka part = total - andar ka window	Complement window
circular array	Array ko double karo ya index % n	Circular window
subsequence / not continuous / negative numbers + sum	Sliding Window NAHI. Prefix Sum + HashMap / DP socho	NOT window

Golden check: **continuous?** + **positive numbers?** + **window ki condition add/remove se update ho sakti hai?**
Teeno YES = Sliding Window.

3. Generic Templates (yaad karne ke liye)

```
# FIXED WINDOW
window = sum(arr[:k]); best = window
for r in range(k, n):
    window += arr[r] - arr[r-k]      # add new, remove old
    best = max(best, window)
```

```
# VARIABLE - LONGEST
l = 0; best = 0
for r in range(n):
    add arr[r] to state
    while state is INVALID:
        remove arr[l] from state; l += 1
    best = max(best, r - l + 1)
```

```
# VARIABLE - SHORTEST
l = 0; best = inf
for r in range(n):
    add arr[r] to state
    while state is VALID:
        best = min(best, r - l + 1)
        remove arr[l] from state; l += 1
```

```
# COUNTING  (subarrays ending at r)
l = 0; count = 0
for r in range(n):
    add arr[r] to state
    while state is INVALID:
        remove arr[l] from state; l += 1
    count += r - l + 1               # all valid subarrays ending at r

# EXACTLY K  =  atMost(K) - atMost(K-1)
```

Dhyan do: 'state' = sum / HashMap count / set / deque - question ki condition ke hisaab se. Yahi har question me badalta hai, baaki template same.

4. Basic Practice (LeetCode me nahi - maine banaye, output ke saath)

Done	#	Question	Level	Priority	Sample input	Output
[ ]	C 1	Max sum of subarray of size K	Easy	MUST	arr=[2,1,5,1,3,2], k=3	9 ([5,1,3])
[ ]	C 2	Count windows of size K with sum >= T	Easy	MUST	arr=[1,2,3,4,5], k=2, T=5	3 (sums: 3,5,7,9)
[ ]	C 3	Min sum of subarray of size K	Easy	SHOULD	arr=[2,1,5,1,3,2], k=3	6 ([1,3,2])
[ ]	C 4	First negative number in every window of size K	Easy	SHOULD	arr=[12,-1,-7,8,-15,30,16,28], k=3	[-1,-1,-7,-15,-15,0] (0 = no negative)
[ ]	C 5	Longest subarray with sum <= K (only positives/zeros)	Medium	MUST	arr=[1,2,1,0,1,1,0], K=4	5 ([1,0,1,1,0])
[ ]	C 6	Longest subarray with at most K distinct numbers	Medium	MUST	arr=[1,2,1,2,3], K=2	4 ([1,2,1,2])

5. LeetCode Questions - Basic to Advanced

Name par click = LeetCode khulega. Stage by stage chalo.

STAGE 1 - FIXED SIZE WINDOW (window ki size K fix hai: ek element add, ek remove)

Done	#	LeetCode	Level	Priority	Twist / kya seekhna hai
[ ]	1	LC 643 Maximum Average Subarray I	Easy	MUST	Same as custom #1, bas sum / K.
[ ]	2	LC 1984 Min Difference Between Highest and Lowest of K Scores	Easy	SHOULD	Pehle sort, phir fixed window - sorting + window combo.
[ ]	3	LC 219 Contains Duplicate II	Easy	MUST	Window ke andar HashSet maintain karna.
[ ]	4	LC 1343 Sub-arrays of Size K and Avg >= Threshold	Medium	MUST	Maximum nahi, COUNT karna hai.
[ ]	5	LC 2090 K Radius Subarray Averages	Medium	SHOULD	Window size = 2K+1, center ke hisaab se answer store.
[ ]	6	LC 1423 Maximum Points You Can Obtain from Cards	Medium	MUST	Reverse thinking: bahar ke cards = andar ka window complement (ya left/right split).
[ ]	7	LC 1052 Grumpy Bookstore Owner	Medium	MUST	Base sum + best window ka extra gain - 2 cheezein combine.
[ ]	8	LC 2461 Max Sum of Distinct Subarrays With Length K	Medium	MUST	Fixed window + HashMap/Set for distinct.
[ ]	9	LC 2134 Min Swaps to Group All 1's Together II	Medium	SHOULD	Circular array: array ko double karo (i % n).
[ ]	10	LC 239 Sliding Window Maximum	Hard	MUST	Window ka max O(1) me chahiye: Monotonic Deque. Most famous.

STAGE 2 - VARIABLE WINDOW, LONGEST (expand right, condition tootne par left shrink)

Done	#	LeetCode	Level	Priority	Twist / kya seekhna hai
[ ]	11	LC 1004 Max Consecutive Ones III	Medium	MUST	'At most K zeros' wali condition. Classic longest-window.
[ ]	12	LC 1493 Longest Subarray of 1's After Deleting One Element	Medium	MUST	Same as 1004 with K=1, answer me -1.
[ ]	13	LC 904 Fruit Into Baskets	Medium	MUST	'At most 2 distinct' - HashMap se count.
[ ]	14	LC 1695 Maximum Erasure Value	Medium	MUST	Unique elements window + sum track.
[ ]	15	LC 2958 Longest Subarray With at Most K Frequency	Medium	SHOULD	Distinct nahi, frequency limit hai.
[ ]	16	LC 2831 Find the Longest Equal Subarray	Medium	SHOULD	Valid window: size - maxFreq <= K (deletions).
[ ]	17	LC 1838 Frequency of the Most Frequent Element	Medium	SHOULD	Sort + window; cost = nums[r]*len - sum.
[ ]	18	LC 2401 Longest Nice Subarray	Medium	OPT	Bitwise AND condition ke saath window.
[ ]	19	LC 1658 Minimum Operations to Reduce X to Zero	Medium	MUST	Reverse: bahar ka sum X = andar ka sum (total - X) ka LONGEST window.

STAGE 3 - VARIABLE WINDOW, SHORTEST / MINIMUM LENGTH (valid hote hi shrink karke minimum nikalo)

Done	#	LeetCode	Level	Priority	Twist / kya seekhna hai
[ ]	20	LC 209 Minimum Size Subarray Sum	Medium	MUST	Longest nahi, SHORTEST. Shrink loop 'while valid' hota hai.

STAGE 4 - COUNTING SUBARRAYS (har right par kitne valid subarrays? / atMost(K) - atMost(K-1))

Done	#	LeetCode	Level	Priority	Twist / kya seekhna hai
[ ]	21	LC 713 Subarray Product Less Than K	Medium	MUST	Har right par count += (right-left+1).
[ ]	22	LC 930 Binary Subarrays With Sum	Medium	MUST	Exactly K = atMost(K) - atMost(K-1) trick.
[ ]	23	LC 1248 Count Number of Nice Subarrays	Medium	MUST	Odd numbers ko 1, even ko 0 maan lo - 930 ban gaya.
[ ]	24	LC 2962 Count Subarrays Where Max Appears at Least K Times	Medium	SHOULD	'At least' type counting.
[ ]	25	LC 2302 Count Subarrays With Score Less Than K	Hard	SHOULD	Score = sum * length; 713 jaisa counting.
[ ]	26	LC 992 Subarrays with K Different Integers	Hard	MUST	Exactly-K distinct = atMost(K) - atMost(K-1). Boss question.

STAGE 5 - ADVANCED / OPTIONAL (jab upar wale solid ho jaaye)

Done	#	LeetCode	Level	Priority	Twist / kya seekhna hai
[ ]	27	LC 862 Shortest Subarray with Sum at Least K	Hard	OPT	Negative numbers: simple window fail. Prefix sum + Monotonic Deque.
[ ]	28	LC 480 Sliding Window Median	Hard	OPT	Window ka median: 2 heaps / sorted list.

LeetCode total: **28** (Easy 3, Medium 20, Hard 5) + 6 custom basics.

6. Company-wise Coding Round Questions (Sliding Window)

Note: Ye official list nahi hai. Ye candidate-reported prep resources (e.g. PapersAdda, 2026) par based hai. Companies exact question repeat nahi karti, par **pattern** repeat hota hai.

Question	Company (reported)	Status
Max Sum Subarray of Size K (C1)	TCS, Wipro, Infosys, Cognizant	Reported
Maximum Average Subarray I (LC 643)	Infosys, Wipro, TCS NQT	Reported
Number of Sub-arrays of Size K and Avg >= Threshold (LC 1343)	TCS / Wipro / Infosys level	Same level, not confirmed
Contains Duplicate II (LC 219)	TCS / Infosys level	Same level, not confirmed
Min Size Subarray Sum (LC 209)	TCS Digital / Infosys level	Same level, not confirmed
Max Consecutive Ones III (LC 1004), Fruit Into Baskets (LC 904)	Infosys SP/DSE, product-company level	Same level, not confirmed
Sliding Window Maximum (LC 239)	Product companies (harder rounds)	Same level, not confirmed

Service companies ke liye kitna kaafi? TCS / Wipro / Infosys me mostly Easy-Medium aate hain. Stage 1, 2, 3 ke MUST questions + custom basics ready ho to kaafi hai. Stage 4-5 product companies aur harder rounds ke liye hain.